



NABH DIGITAL HEALTH STANDARDS FOR HIS/EMR SYSTEMS: ONCOLOGY ANNEXURE

Strengthening Cancer Care through Digital Ecosystems



QUALITY : SAFETY : WELLNESS

Cancer Care and Management Oncology Annexure to NABH Digital Health Standards for HIS/EMR Systems

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The development of the first-ever Oncology-specific Annexure to the NABH Digital Health Standards for Hospital Information Systems (HIS) and Electronic Medical Records (EMR) marks an important milestone in advancing quality cancer care through digital enablement. This initiative reflects NABH's commitment to ensuring that clinical workflows, patient safety, and treatment outcomes in oncology are strengthened by technology-driven standards.

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Jai Hind



Dr. Atul Mohan Kochhar
CEO, NABH

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Introduction

Cancer is one of the leading causes of morbidity and mortality worldwide, and India is witnessing a rapidly growing cancer burden. With increasing incidence, late-stage presentations, and the complexity of oncology care, the disease poses profound challenges for patients, clinicians, and the healthcare system as a whole. Oncology, by its very nature, requires multidisciplinary collaboration, longitudinal care, and integration of advanced diagnostics, therapeutics, and follow-up pathways.

In this context, digital health technologies—particularly HIS (Hospital Information Systems) and EMRs (Electronic Medical Records)—play a pivotal role in streamlining oncology workflows, enhancing clinical decision-making, ensuring continuity of care, and supporting evidence generation. Standardized oncology-specific functionalities within HIS/EMRs are therefore essential to enable clinicians, care teams, and health systems to deliver comprehensive, high-quality, and patient-centered cancer care.

This annexure on Oncology HIS/EMR serves as a specialty certification in addition to the NABH Digital Health Standards for HIS/EMR Systems. It defines oncology-specific capabilities and workflows required for effective and essential cancer care delivery in India, covering aspects such as multidisciplinary tumor boards, treatment planning, chemotherapy and radiotherapy management, survivorship care, and long-term follow-up. By complementing the HIS/EMR certification, this annexure ensures that oncology care providers are equipped with digital systems aligned to the unique needs of oncology practice.

While these standards have been developed through extensive internal deliberations and consultation with oncology experts at National Cancer Grid (NCG) and digital health stakeholders, NABH recognizes that oncology is a rapidly evolving domain. Continuous engagement and feedback from Oncologists, cancer care institutions, digital health companies, and other stakeholders will be critical to refining and updating these standards in alignment with emerging best practices and innovations.

Through this initiative, NABH reaffirms its commitment to advancing excellence in oncology care, strengthening digital health infrastructure, and fostering a more integrated, efficient, and patient-centric cancer care ecosystem in India.

SUMMARY OF THE STANDARDS

Annexure	Standard	Objective Elements	Core	Commitment	Achievement	Excellence
Access Assessment and Continuity of Care (AAC)	1	1	0	1	0	0
Care of Patients (COP)	4	7	1	2	3	1
Management of Medication (MOM)	1	1	0	1	0	0
Information Management System (IMS)	2	2	1	1	0	0
Muti-Disciplinary Care (MDC)	3	7	4	0	0	3
Total	11	18	6	5	3	4

COMPLIANCE LEVEL

Compliance Level for NABH CMS - Oncology Annexure

Category of Objectives	Compliance Level	Total Number of Objectives	Number of objectives to be complied with
Core	100	6	6
Commitment	80	5	4
Achievement	70	3	2
Excellence	50	4	2

ABBREVIATIONS

AAC	Access Assessment and Continuity care
ABDM	Ayushman Bharat Digital Mission
AE	Adverse Event
API	Application Programming Interface
BMI	Body Mass Index
BSA	Body Surface Area
CDSS	Clinical Decision Support System
CMS	Certification of Medical Software
COP	Care of Patients
CPOE	Computerised Provider Order Entry
CTCAE v5.0	Common Terminology Criteria for Adverse Events, Version 5.0
DICOM	Digital Imaging and Communications in Medicine
ECG	Electrocardiogram
EMR	Electronic Medical Record
eMAR	Electronic Medication Administration Record
ENT	Ear, Nose, and Throat
EHR	Electronic Health Record
GFR	Glomerular Filtration Rate
HIS	Hospital Information System
HPR	Histopathology Report
ICD-O / ICD-O3	International Classification of Diseases for Oncology (3rd Edition)

ICMR	Indian Council of Medical Research
IMS	Information Management System
KPI	Key Performance Indicator
LIS	Laboratory Information System
MDT / MDTB	Multi-Disciplinary Team / Multi-Disciplinary Tumour Board
MDC	Multi-Disciplinary Care
MOM	Management of Medication
NCG	National Cancer Grid
NABH	National Accreditation Board for Hospitals & Healthcare Providers
OE	Objective Element
OPD / IPD	Outpatient Department / Inpatient Department
PAC	Pre-Anesthesia Checkup
PACS	Picture Archiving and Communication System
PET-CT	Positron Emission Tomography – Computed Tomography
PS	Performance Status
RIS	Radiology Information System
RT	Radiotherapy
SNOMED CT	Systematized Nomenclature of Medicine – Clinical Terms
TNM	Tumour, Node, Metastasis Staging System
XM CSV / JSON	Data Exchange Formats



ONCOLOGY ENHANCEMENTS TO NABH DIGITAL HEALTH
STANDARDS FOR HIS/EMR SYSTEMS 1ST EDITION

Intent of the annexure

An HIS/EMR system designed for oncology care must capture comprehensive patient information, including demographic details, medical and surgical history, medication records, and laboratory results. Given the chronic and complex nature of cancer, the system should also facilitate the capture of longitudinal data to enable an integrated and coordinated approach to treatment. This oncology-specific annexure builds upon the existing 1st edition NABH Digital Health Standards for HIS/EMR Systems and Objective Elements (OEs) by incorporating oncology-focused enhancements, as well as introducing new standards and OEs tailored to the unique requirements of cancer care and management. The HIS/EMR companies applying for Oncology certification should comply with the enhancements to the existing standards and new standards as per the compliance level mandated by NABH.



Core



Commitment



Achievement



Excellence

CHAPTER 1

Access, Assessment, and Continuity of Care (AAC)



Chapter	Standard	Objective Elements	Core	Commitment	Achievement	Excellence
Access Assessment and Continuity of Care (AAC)	1	1	0	1	0	0

Digital Health Standards and Objective Elements

Summary of Standards Enhanced for Oncology

AAC.3. The system handles laboratory test orders and samples.

Standard

AAC.3. The system handles laboratory test orders and samples.

Objective Elements

Category	Commitment	Head	Common	Type	Functional
Objective Element 3.a	The system configures clinical and administrative workflow for laboratory management.				
Interpretation	Laboratory Management is an important part of a healthcare organization's system, and this system should support the following features: Laboratory Management is vital for healthcare organizations, and the system should support: - Workflows: Setup of master data, process configurations, and information flow between practitioners, the lab, and billing. - Quality Controls: Configuration of essential quality controls. - Report Generation: Ability to generate specialty-specific reports (e.g., microbiology, biochemistry).				

Category	Commitment	Head	Common	Type	Functional
Oncology Enhancement	The system shall support end-to-end integration of the Molecular Biology workflow to enable precision oncology and genomics-informed care. The system shall facilitate molecular-specific sample registration with barcode-based tracking, EMR-enabled test requisition workflows with requisite clinician authorization, and seamless bidirectional integration with the Laboratory Information System (LIS). The system shall support structured bioinformatics reporting for molecular test results, and those reports are integrated into the patient's electronic medical record and linked to their longitudinal health history to support informed clinical decision-making and continuity of care. The system should enable the generation of a unified output that combines molecular biology findings with immunohistochemistry (IHC) and histopathology results to support tumor board discussions and help develop comprehensive treatment plans based on integrated diagnostic data.				



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CHAPTER 2

Care of Patients (COP)



Chapter	Standard	Objective Elements	Core	Commitment	Achievement	Excellence
Care of Patients (COP)	4	7	1	2	3	1

Digital Health Standards and Objective Elements

Summary of Standards Enhanced for Oncology

COP.1.	The system manages consultation services in OPD and IPD
COP.10.	The system supports patient services in remote settings.
COP.12.	The system provides a Clinical Decision Support System.
COP.13.	The system has the capability to create care plans.

Standard

COP.1.	The system manages consultation services in OPD and IPD
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Objective Elements

Category	Commitment	Head	EMR	Type	Functional
Objective Element 1 .a	The system records and reviews initial assessment in OPD and IPD, and patient progress.				
Interpretation	The system shall record and review the initial assessment of patients undergoing treatment in OPD and IPD. During initial assessment, a thorough review is conducted by a designated member of staff. The records of review include vital signs, medical history, physical examination findings, and diagnostic tests, and any previous treatments. The system shall maintain a comprehensive record of all these details. The system should be able to capture initial assessment and re-assessment for different categories of patients, including antenatal, obstetrics, paediatrics, ophthalmology, ENT, etc.				

Category	Commitment	Head	EMR	Type	Functional
Interpretation	The system shall allow nurses and medical practitioners to monitor and analyse the progress of the patients, enabling them to track trends in patients' conditions over time. This feature empowers nurses and medical practitioners to make informed decisions regarding patient care.				
Oncology Enhancement	The system shall also have the functionality to: - Record the list of diagnostic tests that led to confirmation of the primary diagnosis. - Capture and confirm the primary diagnosis along with the date of primary diagnosis and TNM staging of the disease as per clinical guidelines, also capture the modalities used - imaging, histo- cytopathology and any special tests, etc - Document the initial assessment for patients undergoing radiotherapy, including scan details, patient positioning, and immobilisation (e.g., cast) information. - Capture surgery-related data – including details, name, extent, frozen sections, HPR - Capture chemotherapy / radiation related data, including the number of cycles, cycle intervals, date, time, and type of chemotherapy /radiation - indication setting adjuvant, concurrent, neoadjuvant and palliative etc, response / toxicity / any changes (dose medication, incomplete treatment) - Record, view, and update patient details scheduled for chemotherapy, including indications (as defined by the hospital), vitals, and diagnostic results (laboratory and radiology reports/images). - Capture the cause of death occurring during or within 30 days post-completion of treatment, including systemic therapy, radiation therapy, surgery, or concurrent modalities. - Capture and log the identity of the user entering the data. - Refer to Annexure - (A) Surgical Oncology Module ,(B) Radiation oncology Module ,(C) - Chemotherapy Oncology Module				

Category	Commitment	Head	EMR	Type	Functional
Objective Element 1.b	The system provides a summary of the patient's condition, medication order and follow-up visit for the OPD visit				
Interpretation	The system shall generate a concise overview of a patient's health condition, thereby enhancing clinical efficiency and patient care in the OPD. The following shall be included in the overview: Summary of the patient's condition: The Summary could include relevant medical history, current symptoms, diagnosis, and any significant findings from physical examinations or tests. Medication Order Management: The system should provide history and details of the medications being prescribed along with the dosage, frequency, and route of administration, any episode of allergic/ adverse reaction/drug interaction. Follow-Up Visits: The system should facilitate the scheduling and documentation of follow-up visits. This helps healthcare providers and patients to manage upcoming appointments and capture follow-up consultation notes.				
Oncology Enhancements	Follow up: The system shall provide a feature to update the follow-up status (such as active surveillance, stable disease, no evidence of disease, relapsed, referral, any long-term treatment complications), mark appointments and dates of visit planned and the actual outcome (death/survivorship/left treatment)				

Category	Achievement	Head	EMR	Type	Functional
Objective Element 1.c	The system creates order sets based on frequently prescribed medicines				
Interpretation	The system should have a comprehensive medication database (drug names, dosage forms, routes, strengths, indications, contraindications, and potential side effects, as well as the categorisation of medications into different classes or therapeutic categories). This information assists medical practitioners in making informed decisions. Order sets (also known as “abbreviated medication lists”) help to organise and automate the process of placing orders. To facilitate the creation of order sets, the system should offer pre-established templates tailored for common medical conditions or procedures. These templates can then be personalised by medical practitioners to suit their specific needs and the needs of individual patients.				



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Category	Achievement	Head	EMR	Type	Functional
Oncology Enhancement	The system should have a provision to create pre-defined order sets for chemotherapy drugs, selection of brands ,including drug dosage and capping, based on defined pathways or custom-defined as per disease group/organ (as defined by the healthcare organisation) The system should have a provision to modify a chemotherapy drug order set with an option (eg: drop down list) to enter justifications/reasons for modification.				

Category	Core	Head	Common	Type	Technical
Objective Element 1.g	The system has the capability to generate Computerized Provider Order Entry (CPOE) for radiological examinations.				
Interpretation	The system should have the capability to allow medical practitioners to place radiology orders for patients. This functionality empowers medical practitioners to electronically order a diverse array of radiology tests. Computerized Provider Order Entry (CPOE) for radiology services empowers treating medical practitioners with access to the catalogue of available radiology tests. Medical practitioners can select the suitable tests, thereby mitigating potential confusion within both radiology and billing departments, with indications The system should give the schedules of the radiology tests while creating orders. The system may also provide the availability of slots for the use of radiology equipment/appointment list of the radiology department for preparation of patients for example time of fasting and shifting along with provision for special instructions.				
Oncology Enhancement	The system shall have a provision to record details of radiation therapy treatment, including the ability to select the type of procedure, imaging modality (CT, CT+MRI, CT+PET), and pretreatment verification, as well as view and mark checklists for the treatment as specified by the healthcare facility Refer to Annexure B - Radiation oncology Module				

Standard

COP.10.	The system supports patient services in remote settings.
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Objective Elements

Category	Achievement	Head	EMR	Type	Technical
Objective Element 10. b	The system supports effective home care services/ palliative care.				
Interpretation	The system shall assist healthcare organizations to digitally manage home care services. The system should facilitate booking and monitoring of home care services, billing management , and the collection of patient feedback.				
Oncology Enhancement	The system shall support the integration of oncology-specific home care workflows with a focus on patient profiling, adverse event tracking, remote monitoring, medication adherence, teleconsultations, and home nursing management. The system must ensure secure communication, emergency response, and document sharing, with modular capabilities for diet, mental health, and advanced remote diagnostics. The system shall also capture the details of patients (Adult/Paediatric/under the survivorship program as detailed in this document - Refer to Annexure I for Cancer Survivor form and Annexure E for palliative medicine assessment module				

Standard

COP.12.	The system provides a Clinical Decision Support System.
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Objective Elements

Category	Excellence	Head	EMR	Type	Functional
Objective Element 12. a	The system supports the Clinical Decision Support System (CDSS).				
Interpretation	Clinical Decision Support System (CDSS) plays a pivotal role in enhancing clinical decision-making, promoting patient safety, and facilitating effective risk assessment and management within healthcare organizations. By providing evidence-based recommendations, alerts, and guidelines, CDSS tools empower healthcare providers to make informed decisions, thereby improving patient outcomes, reducing medical errors, and enhancing efficiency in care delivery processes. The system shall be equipped to offer CDSS functionality either internally or integrated with external CDSS systems. This functionality can be across a wide range of domains like diagnosis, drug prescriptions, and treatment planning. Refer to Annexure-5 for a non-exhaustive list of common CDSS use cases.				



Category	Excellence	Head	EMR	Type	Functional
Oncology Enhancement	The system shall be able to suggest treatment protocols to doctors based on the diagnosis, staging, pathology reports /molecular pathology reports site, and the host characteristics The system shall trigger an automated alert to the treating clinician in instances where treatment protocol recommendations are unavailable due to incomplete or non-congruent diagnosis, staging, or pathology data. This mechanism ensures proactive identification and flagging of decision-support gaps, thereby upholding clinical governance and patient safety. The system should be able to suggest the dosage based on BSA, AUC, BMI, Height, Weight, Creatine, Age and GFR, etc. (as per defined rules) Refer Annexure H for NCG Clinical Guidelines and protocol				

Standard

COP.13.	The system has the capability to create care plans
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Objective Elements

Category	Achievement	Head	EMR	Type	Functional
Objective Element 13. a	The system has the capability to create customized care plans based on current standards of practice				
Interpretation	The system shall be able to develop customized care plans for specific disease conditions in patients. For example, care plans for individuals with asthma, diabetes, COPD etc. Care plans shall also include elements of assessment and evaluation, goal setting, interventions, monitoring and adjustments, documentation and communication, etc. They may also include aspects of preventive, promotive, curative, rehabilitative, and palliative care.				
Oncology Enhancement	The system shall support the creation of customised care plans based on approved guidelines (e.g, NCG guidelines) or clinician-defined protocols tailored to the organ, disease group, stage, and tumour site. For example, a Care plan for cancer should include elements of a plan for management of comorbidities, a care plan for the cancer, and pain management. For those who have completed treatment, a follow-up plan and survivorship plan are provided. For advanced disease – palliative care It shall maintain a complete audit trail as per the facility's requirements. The system shall allow users to view, update, or modify chemotherapy and radiotherapy plans, including treatment sequence, type, reasons for changes, outcomes, and user remarks.				



Category	Achievement	Head	EMR	Type	Functional
Oncology Enhancement	Additionally, the system shall include a radiotherapy planning requisition module to capture session counts, cycle intervals, planning protocols, site of incidence, region of interest, procedure dates, and diagnostic findings (e.g., PET-CT/CT), in alignment with institutional protocols. Refer to Annexure B for Radiation Oncology Module , Annexure C - Chemotherapy Oncology Module , Annexure E - Palliative Medicine Assessment Module				



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CHAPTER 3

Management of Medication (MOM)



Chapter	Standard	Objective Elements	Core	Commitment	Achievement	Excellence
Management of Medication (MOM)	1	1	0	1	0	0

Digital Health Standards and Objective Elements

Summary of Standards

MOM.3.	The system supports the safe administration of medications
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Standard

MOM.3.	The system supports the safe administration of medications
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Objective Elements

Category	Commitment	Head	EMR	Type	Functional
Objective Element 3.b	The system has the capability of maintaining an electronic medication administration record (eMAR)				
Interpretation	An e MAR provides a comprehensive view of medication administration to the medical practitioners administering medications. The electronic Medication Administration Record (eMAR) system should have the capability to record drugs administered using a specific template. The eMAR should contain: • Dosage: The prescribed amount of the medication. • Route of Administration: The method by which the medication is administered (for example, oral, intravenous, subcutaneous). • Date and Time: When the medication was given. • Administering Personnel: The name or initials of the person who administered the medication and who verified the medication in case of high-risk medications. • Record of any medication administered based on verbal orders.				

Category	Commitment	Head	EMR	Type	Functional
Objective Element 3.b	The system has the capability of maintaining an electronic medication administration record (eMAR)				
Interpretation	An eMAR provides a comprehensive view of medication administration to the medical practitioners administering medications. The electronic Medication Administration Record (eMAR) system should have the capability to record drugs administered using a specific template. The eMAR should contain: Dosage: The prescribed amount of the medication. Route of Administration: The method by which the medication is administered (for example, oral, intravenous, subcutaneous). Date and Time: When the medication was given. Administering Personnel: The name or initials of the person who administered the medication and who verified the medication in case of high-risk medications. Record of any medication administered based on verbal orders.				
Oncology Enhancement	The eMAR should also contain, • Dilution for parenteral with diluent and volume, and rate • Premedication • Post chemo meds – duration, frequency, and name. The system shall have a feature to view all date wise chemotherapy orders of the patient, with status as completed or canceled or modified and reasons (defined drop-down list), if any. The system shall also have a feature to capture date, time and type (e.g. adjuvant) of the planned and actual radiotherapy/ chemotherapy sessions				



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CHAPTER 4

Information Management System (IMS)



Chapter	Standard	Objective Elements	Core	Commitment	Achievement	Excellence
Information Management System (IMS)	2	2	1	1	0	0

Digital Health Standards and Objective Elements

Summary of Standards	
IMS.1.	The system supports healthcare data and interoperability standards for patient, clinical, administrative information to ensure continuity of care, including ABDM
IMS.2.	The system has the capability to support NABH defined key performance indicators and analytical dashboards

Standard

IMS.1.	The system supports healthcare data and interoperability standards for patient, clinical, administrative information to ensure continuity of care, including ABDM
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Objective Elements

Category	Core	Head	EMR	Type	Technical
Objective Element 1.e	The system shall support ICD 10/11 or SNOMED CT covering clinic terminologies for diagnosis, morbidity, and mortality data accurately.				
Interpretation	The system shall support ICD 10/11 or SNOMED CT codes. The system should have the capability to prompt and recommend the relevant ICD 10/11 or SNOMEDCT codes. Implementation of ICD 10/11 or SNOMEDCT can be done through application user interface, backend matching services, or through dedicated medical coding service modules. The system shall support the following coding capabilities: Implement upload, upgrade and deprecation and storage of codes by version into the system				

Category	Core	Head	EMR	Type	Technical
Oncology Enhancement	The system shall also support the capture of ICD-O/ICD- O3 codes to allow the capture of the morphology and grade of the disease. The system shall also support the capture of CTCAE v5.0 codes to enable standardised recording of treatment-related toxicities, including their grade and category				

Standard

IMS.2.	The system has the capability to support NABH defined key performance indicators and analytical dashboards
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Objective Elements

Category	Commitment	Head	Common	Type	Technical
Objective Element 2.a	The system electronically computes and publishes Key Performance Indicators (KPIs) per NABH accreditation standards for hospitals and healthcare organizations				
Interpretation	The system captures relevant patient and administrative data, and computes KPIs as per NABH accreditation standards for hospitals. The system must have the ability to compute the KPIs based on end-user defined periods (start/end dates) and export the KPIs and underlying computation to end-users for further analysis (in JSON, .csv, .xml, .xls, .pdf formats). The list of KPIs to be computed by HIS/EMR system based on NABH accreditation standards is given in Annexure 8 The format for exporting KPI data to NABH is given in the Resource folder on NABH portal. Link to report format				
Oncology Enhancement	The system shall have the capability to capture and share information required for cancer registries as per the requirement in defined formats. For the list of Oncology KPIs to be computed by HIS/EMR system, refer to annexure F1. To understand the NCG Oncology KPI reporting and analytics format, refer to annexure F2				



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CHAPTER 5

Multi-Disciplinary Care (MDC)



Chapter	Standard	Objective Elements	Core	Commitment	Achievement	Excellence
Muti-Disciplinary Care (MDC)	3	7	4	0	0	3

The below set of oncology-specific standards and objective elements have been newly added to ensure a comprehensive oncology HIS/EMR workflow.

Digital Health Standards and Objective Elements

Summary of Standards	
MDC. 1.	The system creates a Multi-Disciplinary Tumor Board workflow
MDC. 2.	The system can capture details related to research and trials
MDC. 3.	The system can capture details related to bone marrow transplantation

Standards

MDC. 1.	The system creates a Multi-Disciplinary Tumor Board workflow
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Objective Elements

Category	Core	Head	HIS	Type	Functional
MDC 1.a	The system enables patient selection, tumor board ID assignment, MDTB review scheduling, and automated clinician notifications.				
Interpretation	A Multi-Disciplinary Tumor Board (MDTB) is a collaborative team of oncologists, radiologists, pathologists, surgeons, and other specialists who review complex cancer cases to ensure optimal treatment planning as per the standard treatment guidelines with individualization based on the patient and disease characteristics. Integrating MDTB into HIS/EMR enhances patient care by enabling structured case discussions, decision documentation, and evidence-based treatment coordination, improving outcomes and efficiency.				

Category	Core	Head	HIS	Type	Functional
Interpretation	The system should allow the treating medical practitioner to select a patient for MDTB based on configurable clinical criteria such as confirmed or possible diagnosis, TNM stage, and organ system. It should also enable the selection of required medical practitioners for the MDTB and send appointment notifications to block their calendars.				

Category	Core	Head	EMR	Type	Functional
MDC 1.b	The system displays the list of patients scheduled for MDTB meetings, allowing the medical practitioner to select and view patient-specific details on a single screen.				
Interpretation	The system should provide a centralized, dynamic list of patients scheduled for upcoming tumor board meetings, with filters by date, specialty, or disease category. Users should be able to select a patient and review all relevant clinical data—case notes, lab results, procedure details, radiology/pathology images, and supporting documents—in a single screen. Patient data should be seamlessly pulled from EMR, PACS, LIS, and RIS to enable comprehensive, streamlined access.				

Category	Core	Head	HIS	Type	Functional
MDC 1. c	The system records the attendance and details of medical practitioners who attend the MDTB meeting, and documents them as per the department.				
Interpretation	The system shall allow authorised users to view the list of medical practitioners assigned to the MDTB meetings. The system should have an MDTB attendance feature that records the details of medical practitioners who attended the meeting, such as name, designation, speciality, and role in the proceedings, ensuring accountability and compliance with multidisciplinary team (MDT) documentation standards. The system could facilitate the sending of automated calendar reminders to medical practitioners participating in tumor board discussions to ensure timely participation.				



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Category	Excellence	Head	EMR	Type	Functional
MDC 1.d	The system has the capability to document a summary of the tumour board discussion related to the diagnoses and treatment recommendations.				
Interpretation	The system should document the individual inputs related to diagnosis, treatment recommendations, and follow-ups, and capture a summary of the multidisciplinary inputs in a standardized format. Refer Annexure D for multidisciplinary module				

Standards

MDC. 2.	The system can capture details related to research and trials
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Objective Elements

Category	Core	Head	EMR	Type	Functional
MDC 2.a	The system has the provision to capture consent, label patients enrolled under research trials, and view the list of research trials conducted.				
Interpretation	The system provides a provision to assign patients to research trials by using a check box. The system shall have a provision to capture the following, Screened for trial - Yes/No..... Eligible for trial - Yes/No..... Consent given yes/No , - Ability to upload the consent The system shall have ability to capture other details e.g., Ethic's Committee Trial Name..... Trial ID The system shall also have a provision to view the list of patients enrolled in the past/present for ongoing/past research and clinical trials conducted. Refer Annexure J for Sample consent form for Clinical Trail				

Category	Excellence	Head	HIS	Type	Functional
MDC 2.b	The system supports the end-to-end management of clinical and research trials, including randomization and enrollment, visit tracking, investigational product management, adverse event monitoring, and close-out and archival				

Category	Excellence	Head	HIS	Type	Functional
Interpretation	The system shall provide structured functionalities to manage all key components of a research or clinical trial, ensuring traceability and regulatory compliance. Specifically, the system shall include: Randomization & Enrollment: The system shall support randomization logic and maintain logs of enrolled participants with their trial arms. Visit Management: The system shall allow configuration and scheduling of study-specific visits and procedures, and enable tracking of visit adherence and completion status. Investigational Product (IP) Management: The system shall provide inventory management for investigational drugs/devices, including batch tracking, dispensing logs, and accountability records. Adverse Event Monitoring: The system shall enable documentation and classification of adverse events (AEs), their severity, causality assessment, and linkage to trial protocols. Close-Out & Archival: The system shall facilitate closure of trial records with final status, ensure archiving of all related documents (including approvals, monitoring reports, and logs), and provide controlled access for audits or regulatory review. The system should maintain linkage of this module with patient master data, consent records, ethics committee approvals, and ensure role-based access to maintain confidentiality and integrity of research data				

Standards

MDC. 3.	The system can capture details related to bone marrow transplantation
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Objective Elements

Category	Excellence	Head	EMR	Type	Functional
MDC 3.a	The system has the provision to enter donor and recipient details in defined templates for bone marrow transplant.				
Interpretation	The system shall have the provision to enter donor and recipient details in defined templates (fields as defined by the facility) for bone marrow transplants. The transplant will be both Autologous and Allogeneic transplants				

Category	Excellence	Head	EMR	Type	Functional
Interpretation	Autologous transplant should capture the following, 1. Indication 2. Disease status at transplant 3. Pre-transplant work up – PS, ECHO, ECG, Dental, PFT, GFR any other referrals 4. Stem cell mobilization agents 5. Conditioning therapy and dates 6. Date of stem cell infusion 7. Stem cell dose 8. Date of Neutrophil Engraftment 9. Date of platelet engraftment 10. Any significant complication				
	Allogeneic Transplants should capture the following, 1. Indication 2. Disease status at transplant 3. Pre-transplant work up – PS, ECHO, ECG, Dental, PFT, GFR any other referrals 4. Type of donor – MSD/Haplo/ MUD/ others 5. Donor workup details 6. Conditioning therapy and dates 7. Date of stem cell infusion 8. Stem cell dose 9. Date of Neutrophil Engraftment 10. Date of platelet engraftment 11. GVHD prophylaxis 12. Acute GVHD details 13. Chronic GVHD details 14. Treatment of GVHD 15. Any significant complication				
	The system shall also have provisions, 1. To process internal approvals for the transplantation procedure (E.g., transplant committee approval) 2. There is a provision for sharing reports with authorized agencies (reports such as the consent form, Psychiatric evaluation, doctor's recommendation, etc.).Refer Annexure G1 and G2 for consent forms				



Core



Commitment



Achievement



Excellence

ANNEXURE

- A) Surgical Oncology Module: <https://www.kcdo.in/src/docx/ner-surgical-oncology-module-2.0.pdf>
- B) Radiation Oncology Module: <https://www.kcdo.in/src/docx/ner-radiation-oncology-module-2.0.pdf>
- C) Chemotherapy Oncology Module: <https://www.kcdo.in/src/docx/ner-medical-oncology-module-2.0.pdf>
- D) Multidisciplinary Tumour Board Module: <https://www.kcdo.in/src/docx/ner-multi-disciplinary-tumor-board-module-2.0.pdf>
- E) Palliative Medicine Assessment Module: <https://www.kcdo.in/src/docx/ner-palliative-medicine-assessment-module-2.0.pdf>
- F) 1) NABH Key Performance Indicators: Cancer Care and Management (Page 21 – 27)
2) NCG KPI Sample Tabular Format: https://docs.google.com/document/d/1fzH-5J_Cd4Awi2ENxn0_TKBTllpfc5wm/edit#heading=h.8nomxu9t5981
- G) 1) Transplant consent: consent form sample template
2) Transplant committee approval template
- H) NCG Clinical Guideline and protocol - <https://www.ncgindia.org/cancer-guidelines> (2019)
- I) Cancer survivor form: Cancer Survivor form
- J) Clinical trial sample consent form - https://actrec.gov.in/sites/default/files/TMC_IEC/3.%20Template%20for%20Informed%20Consent%20Form.pdf

F1 : Key Performance Indicators : Cancer Care Management

S. No.	Indicator	Definition	Inclusion criteria	Exclusion criteria	Formula	Unit	Duration of Reporting	Data Variables	HIS/EMR Guide
1	Cancer Case Load	Measures the number of cancer cases confirmed through histopathological/cytopathology examination during a defined reporting period.	All cancer cases confirmed by histopathology and cytopathology	Provisionally diagnosed or non-confirmed cases (e.g., clinical, radiological, or suspected without pathological confirmation).	$\frac{\text{(Number of cancer cases confirmed by histopathology/cytopathology during the defined timeframe)}}{\text{(Total number of cancer patients registered during the same timeframe)}} \times 100$	%	Daily / Monthly / Bi-annually / Annually	Histopathology reports / Cytopathology reports	Capture biopsy/cytology details, structured pathology reports, confirmation status- "Confirmed by histopathology/cytopathology" (Yes/No)
2	Distribution of cancer cases by relevant stage at final diagnosis	Distribution of cancer cases by stage group at final diagnosis.	All individuals with confirmed cancer diagnosis within the defined timeframe	Individuals without a final diagnosis or staging	$\frac{\text{(Number of cases at each stage group)}}{\text{Total number of finally diagnosed cancer cases}} \times 100$	%	Daily / Monthly / Quarterly / Annually	TNM staging	Use most recent staging data at the time of final diagnosis
3	Percentage of cancer surgeries performed Vs advised	Measures the proportion of patients who underwent surgery within 3 months of being advised	Patients advised surgery and subsequently operated within 3 months	Patients not advised surgery	$\frac{\text{(Number of surgeries performed within 3 months of advice)}}{\text{Total patients advised surgery in the MDT plan}} \times 100$	%	Daily to Annually	MDT Plan, Name of procedure	Map treatment plan decisions (e.g., from MDT discussions) to surgery dates. Backend capture of alternate treatments or deviations from planned surgery.

S. No.	Indicator	Definition	Inclusion criteria	Exclusion criteria	Formula	Unit	Duration of Reporting	Data Variables	HIS/EMR Guide
4	Treatment modalities according to clinical extent of disease	Treatment modalities planned in MDT Surgery only: Treatment involving surgical procedures alone. Systemic therapy only: Treatment involving systemic therapy alone. Radiotherapy only: Treatment involving radiotherapy alone. S+R: Combined treatment involving surgery and radiotherapy. S+C: Combined treatment involving surgery and systemic therapy. R+C: Combined treatment involving radiotherapy and systemic therapy. S+R+C: Combined treatment involving surgery, radiotherapy, and systemic therapy. Symptomatic care: Supportive care focused on managing symptoms without curative intent.	All individuals with a confirmed diagnosis of cancer. Cases with complete information on treatment modalities used	Cases with missing or incomplete information on treatment modalities.	$\left(\frac{\text{Number of Cases in Each Treatment Modality Category as per the treatment plan in MDT}}{\text{Total Number of Cancer Cases in MDT}} \right) \times 100$	%	Monthly / Quarterly / Annually	Treatment plan in MDT Module	Classify S/R/C combinations. Auto-capture from MDT plan. Maintain temporal history.
5	Mortality Rate	Captures mortality data for cancer patients in hospital	All confirmed cancer deaths occurring in the hospital	Deaths occurring outside the hospital setting	$\left(\frac{\text{Number of inpatient deaths}}{\text{Total cancer-related admissions}} \right) \times 100$	%	Monthly to Annually	Cause of death, Date of admission, Reason for admission, Type of surgery, 30/90-day mortality, date of last therapy before death	Capture death as surgery outcome. Record cause of death, Capture reason for admission and procedure details

S. No.	Indicator	Definition	Inclusion criteria	Exclusion criteria	Formula	Unit	Duration of Reporting	Data Variables	HIS/EMR Guide
4	Treatment modalities according to clinical extent of disease	Treatment modalities planned in MDT Surgery only: Treatment involving surgical procedures alone. Systemic therapy only: Treatment involving systemic therapy alone. Radiotherapy only: Treatment involving radiotherapy alone. S+R: Combined treatment involving surgery and radiotherapy. S+C: Combined treatment involving surgery and systemic therapy. R+C: Combined treatment involving radiotherapy and systemic therapy. S+R+C: Combined treatment involving surgery, radiotherapy, and systemic therapy. Symptomatic care: Supportive care focused on managing symptoms without curative intent.	All individuals with a confirmed diagnosis of cancer. Cases with complete information on treatment modalities used	Cases with missing or incomplete information on treatment modalities.	$\left(\frac{\text{Number of Cases in Each Treatment Modality Category as per the treatment plan in MDT}}{\text{Total Number of Cancer Cases in MDT}} \right) \times 100$	%	Monthly / Quarterly / Annually	Treatment plan in MDT Module	Classify S/R/C combinations. Auto-capture from MDT plan. Maintain temporal history.
5	Mortality Rate	Captures mortality data for cancer patients in hospital	All confirmed cancer deaths occurring in the hospital	Deaths occurring outside the hospital setting	$\left(\frac{\text{Number of inpatient deaths}}{\text{Total cancer-related admissions}} \right) \times 100$	%	Monthly to Annually	Cause of death, Date of admission, Reason for admission, Type of surgery, 30/90-day mortality, date of last therapy before death	Capture death as surgery outcome. Record cause of death, Capture reason for admission and procedure details

S. No.	Indicator	Definition	Inclusion criteria	Exclusion criteria	Formula	Unit	Duration of Reporting	Data Variables	HIS/EMR Guide
6	Systemic therapy Completion Rate	Measures percentage of patients completing planned systemic therapy.	Patients who initiated systemic therapy, Patients who completed planned systemic therapy as per treatment plan	Patients for whom treatment was planned in the MDT but not initiated. Patients who opted for an alternative treatment plan after discussion with healthcare providers.	$\left(\frac{\text{Number of patients completing planned systemic therapy}}{\text{Total patients Initiated on systemic therapy}} \right) \times 100$	%	Daily / Monthly / Quarterly / Annually	Systemic therapy status, Completion status, Discharge Summary at completion / Reason for Non-Completion	Discharge summary at completion to have a status update asking the clinician to fill- completed planned no of cycles- Yes/No, if no specify the reasons from the drop-down list- Death/Progression/de faulter/Tolerance/ Others.
7	Radiation Therapy Completion Rate	Percentage of patients who successfully complete their prescribed radiation therapy	All Cancer patients for whom radiation therapy was part of the treatment plan as a part of MDT treatment planning Patients who successfully completed the prescribed course of radiation therapy within the specified timeframe	Patients for whom the radiation therapy plan was modified due to medical reasons Patients who opted for an alternative treatment plan after discussion with healthcare providers	$\left(\frac{\text{Number of patients completing radiation therapy}}{\text{Total patients Initiated on radiation therapy}} \right) \times 100$	%	Daily / Monthly / Quarterly / Annually	Radiation therapy completion- Planned or unplanned Reason for Gap- Poor Compliance/ Treatment related toxicity/ Machine breakdown/ Others Type of Radiation therapy- EBRT, Brachytherapy, Both	Type of radiation therapy- EBRT/Brachytherapy/ Both EBRT- Radiation therapy completion date, Treatment completion- Planned or Unplanned, Treatment Gap- Yes/No, Reason for Gap- Poor Compliance, Treatment related toxicity, Machine breakdown, Others In case of Brachy- it is Total Radiotherapy Dose planned Vs given at the time of completion, Total no of fractions/Fractionation delivered date and time/ Dose per fraction.

S. No.	Indicator	Definition	Inclusion criteria	Exclusion criteria	Formula	Unit	Duration of Reporting	Data Variables	HIS/EMR Guide
8	Average Time from Presentation /Screening to Final Cancer Diagnosis	Median time from initial presentation or screening to confirmation of final cancer diagnosis.	All newly diagnosed cancer cases where time is measured from first hospital registration or screening	Pre-diagnosed cases who have been partially treated outside	Median of time (in days) from presentation/screening to confirmed diagnosis for all newly diagnosed cancer cases.	Days	Monthly / Quarterly / Annually	Registration date, Histopath date	Histopath reporting date, date of registration, Auto-calculate time difference and tag TAT category in structured field
9	Median Time from Diagnosis to Initiation of Definitive Treatment for Cancer	Median time from confirmed cancer diagnosis to initiation of definitive treatment.	Cases that received surgery, chemotherapy, or radiation as first definitive treatment	Cases with missing or unreliable data regarding the time of diagnosis or initiation of treatment. Cases where treatment initiation was delayed for reasons unrelated to the hospital's processes (e.g., patient preference, comorbidities).	Median of time intervals (in days) from diagnosis to start of first definitive treatment across all eligible cases.	Days	Monthly / Quarterly / Annually	Histopathology reporting date, Protocol assignment date/ Surgery date	Histopathology reporting date, Treatment initiation dates- maintain a temporal history of the patient journey.
10	Relapse after treatment	First recurrence within 5 years post-treatment for non-metastatic cancer.	Patients diagnosed with non-metastatic (Stage I-III) solid tumors or hematologic cancers.	Missing/incomplete recurrence data Patients presenting with metastatic disease at diagnosis (Stage IV). Patients who	$(\text{Patients with Recurrence} / \text{Total non-metastatic patients completed treatment 5 years ago}) \times 100$	%	Annually	Gender, Age, State, Stage, Modality, Recurrence Status	Use Decision Panel. Track end-of-treatment and recurrence status. Use mapped surgical codes.

S. No.	Indicator	Definition	Inclusion criteria	Exclusion criteria	Formula	Unit	Duration of Reporting	Data Variables	HIS/EMR Guide
10	Relapse after treatment	First recurrence within 5 years post-treatment for non-metastatic cancer.	Patients who have completed curative-intent treatment (e.g., surgery, definitive chemoradiation).	received only palliative treatment. Patients lost to follow-up immediately post-treatment (within 3 months). Patients with second primary malignancy within the 5-year follow-up window. Cases where recurrence cannot be reliably determined due to incomplete documentation or lack of imaging/pathological confirmation.	Numerator: Number of patients with non-metastatic cancer who experienced a documented first recurrence (local, regional, or distant) within 5 years after completion of primary curative-intent treatment.	%	Annually	Gender, Age, State, Stage, Modality, Recurrence Status	Use Decision Panel. Track end-of-treatment and recurrence status. Use mapped surgical codes.
11	Hospital Length of Stay	Average hospital stay duration for surgical or inpatient treatment.	Hospital stay duration calculated from admission to discharge Surgical procedures (elective or emergency) Inpatient treatments related to cancer (e.g., complications, planned admissions for chemo/radiotherapy)	Admissions unrelated to cancer (e.g., comorbid condition management) Day-care or outpatient procedures not requiring overnight admission	Total number of hospital days ÷ Total number of patients undergoing surgery or inpatient care	Days	Monthly / Quarterly / Annually	Type of Procedure/ Admission/Discharge dates, Treatment modality, post-op monitoring- Routine/High dependency/ Intensive care, Clavien Dindo grade-1/2/3/4/5	Track admission/discharge, reason for admission-surgery or inpatient treatment, level of post op monitoring from anesthesia module marked as Routine/High dependency/ Intensive care, Post-op complications, list of surgeries/procedures mapped to OT scheduler, Clavien Dindo grading of worst complication-1/2/3/4/5

S. No.	Indicator	Definition	Inclusion criteria	Exclusion criteria	Formula	Unit	Duration of Reporting	Data Variables	HIS/EMR Guide
12	Palliative Care Referral	Percentage of patients referred to palliative care.	All patients for whom treatment intent was marked as palliative	None	(No of pts where intent was palliative and were referred to palliative care/ Total no of pts whose intent was recorded as palliative) *100	%	Monthly / Quarterly / Annually	Referral date, Department referred to, Stage, Intent	Capture referral date and department, Intent to treatment as curative and palliative, Auto-flag non-curative patients who were not referred to palliative care.

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